

Practice Session 2

```
library(SDS1000)
```

Part 1: Measures of central tendency for quantitative data

1.1 Calculating the Sample Standard Deviation by Hand

Here is the formula for the sample mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

And here is the formula for the sample standard deviation:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Using the following data, complete the table for calculating the sample standard deviation.

Cost(Dollars)	b. Deviations: $x_i - \bar{x}$	c. Deviations squared: $(x_i - \bar{x})^2$
850		
900		
1400		
1200		
1050		
750		
1250		
1050		
565		
1000		

a. The mean of the costs:

$$\bar{x} =$$

b. Write the sum of squared deviations value here:

$$\sum_{i=1}^n (x_i - \bar{x})^2 =$$

c. Write the value of the squared deviations divided by $n - 1$ here:

$$\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} =$$

d. Write the value for the square root of the sum of squared deviations here:

$$s = \sqrt{\dots} =$$

1.2 Five-Number Summary

Using the numbers from the previous exercise, do the following:

- Find the 5-number summary (minimum, Q1, median, Q3, maximum), you can use the function `fivenum()`.
- Check your work using R functions

Answers

```
# your code here
```

1.3 Boxplots and Histograms

Consider the `mtcars` data set. This dataset is built into R, so you can access it directly; no downloads required!

You can learn more about this dataset using `?` in the console or the `View()` function. You can also use the functions `head()` and `tail()`.

1. First, create a histogram of the variable `mpg`. Then, create a boxplot of `mpg`. How do these two plots compare?
2. What proportion of cars have fuel consumption less than(or equal to) 15 mpg?
3. What proportion of cars have fuel consumption less than(or equal to) 35 mpg?
4. Create a boxplot of `mpg` by number of cylinders `cyl`.

Answer

```
#1# Create a histogram of 'mpg'
# your code here

## Create a boxplot of 'mpg':
# your code here

#2 proportion of cars have a fuel consumption of less than(or equal to) 15 mpg
# your code here

#3 proportion of cars have a fuel consumption of less than(or equal to)35 mpg
# your code here

#4 create boxplot of mpg per cylinder:
# your code here
```

The histogram shows the overall shape and spread of `mpg`, revealing that the distribution is slightly right-skewed with most cars between about 15 and 25 mpg. The boxplot summarizes the same data with the five-number summary (median, quartiles, and whiskers), making it

easier to quickly identify the center, spread, and any potential outliers, but it does not show the shape (e.g., skewness or multiple peaks) as clearly as the histogram does.

1.4 Quantitative data : histograms and outliers

Generate histograms for each of the following data sets. Use the `$` command to access the individual data sets.

1. For each histogram, add the mean to the plot using `abline()`.
2. Add the median using `abline()`.
3. Do you see any potential outliers? Also calculate the five-number summary for each using R.

```
set.seed(999)
data = data.frame(
  dat1 = -rchisq(1000, df = 1),
  dat2 = rchisq(1000, df = 1),
  dat3 = runif(1000),
  dat4 = rnorm(1000),
  dat5 = sample(c(rnorm(1000, mean = 2), rnorm(1000, mean = 10)), size = 1000)
)
```

Answers:

```
# your code here
```

1.5 Percentiles

Compute the 25th, 50th, and 75th percentiles for the 5 datasets in the `data` data.frame. Which has the smallest median? Which has the largest? You can use the function `quantile()`.

```
# data 1
# your code here

# data 2
# your code here

# data 3
# your code here

# data 4
# your code here

# data 5
# your code here
```

Part 2: Normal Distribution and ± 2 Standard Deviations

2.1 Example:

The normal distribution (also known as the “bell-curve”) occurs very frequently in mathematics, statistics, and the natural and social sciences.

Which of the 5 data sets in the `data` data.frame appears to be normally distributed?

Answers:

```
#par(mfrow=c(3,2))

# your code here
```

2.2 Calculate the endpoints of one standard deviation above and below the mean.

Using this data set, find the mean and standard deviation, then calculate the endpoints of **1 standard deviation above**, and **1 standard deviation below the mean**. What percentiles do these values correspond to?

Note: Approximately 68% of the data lie between 1 standard deviation above and 1 standard deviation below the mean.

Answers

```
# your code here
```

2.3 z-Scores

Expressing a distance from the mean in standard deviations standardizes the performance. To **standardize** a value, we subtract the mean and then divide this difference by the standard deviation:

$$z = \frac{y - \bar{y}}{s}$$

The values are called **standardized values**, and are commonly denoted with the letter z . Usually we just call them **z-scores**.

z -scores measure the distance of a value from the mean in standard deviations. A z -score of 2 says that a data value is two standard deviations above the mean. It doesn't matter whether the original variable was measured in fathoms, dollars, or carats; those units don't apply to z -scores.

Data values below the mean have negative z -scores, so a z -score of -1.6 means that the data value is 1.6 standard deviations below the mean.

Of course, regardless of the units, the farther a z -score is from 0, the more unusual it is. Thus, a z -score of -1.3 is more extraordinary than a z -score of 1.2.

2.4 Example of comparing z-Scores

A town's January high temperatures average 2°C with a standard deviation of 6° , while in July the mean high temperature is 24°C and the standard deviation is 5° .

In which month is it more unusual to have a day with a high temperature of 13°C ? Explain.

Answers:

```
# January *z*-score
# your code here

# July *z*-score
# your code here
```